

Subsystem Gap Analysis

PREPARED BY Energy Hippo implementation team

For each EEMSuite subsystem: **what the product supports, what Northlake has today, the gap, and what to add**. Grounded in energyhippo/EEMSuite code, the live EEMSuite DB, and the current dataset in this repo.

Legend: ● covered · ● partial · ● missing.

0. Foundation: hierarchy, security, units, calendar ●

What EEMSuite needs. Almost every one of the 94 data-backed reports starts from a spine: the org tree (MetaWorld_TBL / MetaLevel_TBL), security (ValidUser, Users_Tbl, UserGroup*, MetaWorldPermissions_TBL, Permissions_TBL, SiEAppObjects), units (Unit, UnitProfile, MeasureType, UnitConversion), a fiscal calendar (CompanyCalendar), and meters tied to billing accounts (Meter, BillingAccount_Tbl, UtilityProvider). **If the substrate is not loaded with a permissioned user, every report returns empty.**

Permission model: access is gated three ways — (a) **which hierarchy nodes** a user sees (MetaWorldPermissions_TBL, type D Points permissions, recursive down the tree), (b) **which screens/reports** (SiEAppObjects + Permissions_TBL types \$/A/D/EX/F/MI/MU/TS), and (c) **authority level** (AccessLevel 1 read / 2 edit / 3 approve — e.g. bill QC approval requires ≥3).

Northlake today. Strong **design**: 3 EEM companies sharing the same hierarchy levels (Site, Building, Meter, Point), 19 users, 27 groups, 16 permission profiles, disabled-user case. Units include the unusual ones (ton-hours, klb, ERU). All in YAML, **not loaded**, and no built pipeline to load it.

Gap. (1) No generate → load pipeline that populates an EEMSuite DB. (2) Permission **profiles** are semantic intent, not yet mapped to concrete Permissions_TBL/SiEAppObjects grants. (3) CompanyCalendar periods not materialized.

What to add. A Phase-0 loader; concrete permission grants per group (incl. node-scoped D grants proving a Finance user sees only some buildings, and an AccessLevel-3 approver vs. a viewer); a 2024–2025 monthly CompanyCalendar with **1–2 months deliberately omitted** to exercise the "fiscal period not found" import rejection.

1. Monthly bills + charges ●

What EEMSuite needs. Bill + BillItem (line items) routed into a BillScenario ("Actual"), ≥13 consecutive months per account so variance and year-over-year tests bind. Reports: AccountHistory, BillHistory, CostBreakOut, BillVariance(+Analysis/Detail/Ranking), FacilityAccountDetail, PerformanceSnapshot, FiscalPeriodBills, MonthlyUtilityDistribution, etc.

Northlake today. Bills are modeled at the account level (24 months, all providers) with usage/demand/total — good coverage of **bill-level** facts and the estimated → corrected, meter-replacement, account-renewal, duplicate-import, and rate-change scenarios.

Gap. The `bill` source entity has `usage/demand/total_cost` but **no charge-line breakdown** (`BillItems`). `CostBreakOut`, `BillingCharges`, `RateComponent` mapping, and GL coding all need itemized lines (energy vs demand vs fixed vs tax).

What to add. Extend the `bill` model with charge lines (commodity, determinant linkage, quantity, unit, amount, GL account) so each bill decomposes into the same components the rate engine would produce.

2. Interval / time-series use •

What EEMSuite needs. Points (`PointDef_TBL`) with rows in `AnalogArchive_TBL` (`PtID` × `UTCTimeStamp` × `Interval`, `Value/Min/Max/Avg`) and `DigitalArchive_TBL` for status points. Drives all 15 Use Analysis reports (`TimeInterval`, `24HourLineChart`, `MultiPointTrend`, `HeatMap`, `LoadDurationCurve`, `Histogram`, `ScatterPlot`, `Summary`, `AverageHourlyProfile`, `DigitalSummary`, `DataViewExport`, `MonthlyEUI`, `UsageVariance`, `AggregateDemand`, `AggregatePeakLoad`).

Northlake today. Strong: 15-min electric (Science Center, Student Center, Lakeview Residence Hall, Recreation and Aquatics Center, Central Plant) and solar (3 arrays), hourly chilled water and Central Plant production, hourly apartment electric submeters, 5-min chiller kW, weather/sensor series.

Gap. Few **digital/status** points (`DigitalSummary` stays thin); interval coverage is concentrated on a few meters.

What to add. A handful of digital points (e.g. chiller enable/occupancy status) with `DigitalArchive` data; optionally widen interval coverage so load-duration/heat-map reports have variety. (Mostly already covered.)

3. Rate engine • → the biggest "supported but unexercised" gap

What EEMSuite supports. A rate = `UtilityRate` → rate classes (`UtilityRateClass`) → determinants (`UtilityDeterminant`). **The determinant is the charge.** There are exactly **9 math kinds** (each a `UtilityRates_Run_NNN` proc):

Type	Kind	Northlake has it?
100	RTP / real-time index pricing (hourly price point)	• no
200	Charge period: energy/demand, tiered/block	• energy/demand/tiers described
210	Stepped fixed-value	• no
300	Fixed by day	• (water meter charge)
400	Fixed by period (customer/facility charge)	• (facility charge)
500	Minimum bill	• no
600	Maximum bill / cap	• no
700	Bill adjustment (% rider)	• not at determinant level
800	Taxes (% on subtotal, compounding)	• not at determinant level

Plus cross-cutting constructs: **time-of-use** (one determinant per period via schedules), **seasonal** (summer/winter via `UtilitySeason`), **holiday calendars**, **ratchet demand** (`UtilityDeterminantRatchet`), **load factor**, **rate-class options** (voltage/phase/tax-exempt/farm), **pre/post determinant scripts**,

effective-dated proration / escalation, unit conversion, multi-currency, and net-metering credits

(modeled as negative blocks). The engine does **not** natively support: power-factor math, coincident-peak/4-CP, CPP event pricing, or net-metering banking — those must be approximated.

In the live demo DB, types **100 (RTP)**, **210 (stepped)**, **600 (max-bill)** have **zero** determinants, and **ratchets** have only 1 row — so these are the constructs least proven anywhere.

Northlake today. Rate **metadata** for TOU energy, demand, facility/customer charge, tiered water, seasonal gas, volumetric sewer, flat-per-ERU stormwater, PPA energy, internal allocation, and a FY2025 rate change. Described at bill level, not modeled to the determinant/block/schedule level the engine consumes. The `rate_schedule` source entity has no field block yet.

Gap. No determinant-level model; the exotic kinds (RTP, stepped, max-bill, ratchet, load-factor, %-rider, tax, min-bill, class options, scripts, multi-currency) are entirely absent.

What to add. Grow the rate model to determinants/blocks/schedules and author **~10 rate schedules** that, between them, fire all 9 `Run_NNN` procs and every construct — e.g. residential inclining-block (200/400/500/800), commercial TOU+seasonal demand with holidays + riders (the flagship), large-demand ratchet + load factor, base/excess contract demand with scripts, **RTP index** (the only way to fire type 100; needs an hourly price point), **stepped + max-bill** (types 210/600), gas with monthly fuel-adjustment proration (300 + effective dating), multi-commodity water/sewer/CHW/steam, and a rate-class-options demo. Each rate must be **assigned to meters** (`UtilityMeterRates`) with ≥ 13 months of usage so ratchets/seasons/proration bind. (Full list in [roadmap.pdf](#) Phase 2.)

4. AP / GL / financial interface •

What EEMSuite supports. A config-driven export engine. `ExportSetup` names per-format procs (Approval/UploadFile/Batch); the file layout lives in the "UploadFile" builder proc. Export eligibility gates on `BillingAccount.Type_Code` (0 none / 1 AP / 2 GL / 3 both) and active status. GL coding: `GLaccount + 5` user-defined segments (`GLaccountUDField/Value`), attached to bills via `BillItem.BI_ExpAcctID` (and account defaults). **7+ distinct real export file formats** exist in the codebase:

1. Single-record fixed-width AP (`export_ap`, .txt)
2. Fixed-width-in-CSV AP (`export_ap2`, .csv)
3. Header+detail fixed-width AP with GL-UDF coding (Sacramento County — region matches Northlake)
4. Multi-record (C/H/L/D) fixed-width AP voucher (Massport / PeopleSoft)
5. Multi-record (H/L) fixed-width GL journal with balanced debit/credit pairs (Massport / PeopleSoft)
6. Comma-delimited SAP FI document (Clark County)
7. AR interface (Massport)

Reports: `AccountsPayableDetail`, `APApprovalAndUpload` (Financial Interface Approval), `GLAccountUDV`. Bill lifecycle states tracked in `StatusHistory` (Received → Bill Entry → QC Approval → Payment Approval → ... → G/L Upload).

Northlake today. Nothing — GL coding was explicitly deferred in Stage 1.

Gap. No GL accounts, no UDF segments, no `ExportSetup` rows, no export runs, no bill GL coding, no approval/upload flow.

What to add. A chart of expense/AP GL accounts per company; the 5 UD segments (Order/FTC/Cost Center/Fund Center/Fund or similar) with realistic + partly-blank coverage; bill `BI_ExpAcctID` coding;

ExportSetup rows + builder procs so we can **generate one file of each of the 7 formats**; mixed Type_Code accounts (incl. a 0/no-upload negative control); approved-vs-blocked bills (one blocked by an open validation alert). This directly answers "create actual AP interface files."

5. Tenant rebilling & allocation •

What EEMSuite supports. Two top-level modes — **Bill Generation** (rate × meter use → recalculated tenant bill) and **Bill Allocation** (split a parent bill's cost to tenants) — plus a lower-level **WUI point allocation** engine (disaggregate a master point's use to sub-points). **7 bill-allocation methods** (verified): Fixed Amount & %, Fully Submetered, Partially Submetered, Building Areas, Partially Submetered w/ Coincident Demand, **Bill Aggregation** (children → parent), Metered Substations. **3 point-allocation methods**: Weighted Use Index, Measured Use, and **Percentage Based (catalog only — unimplemented; document as a known gap)**. Bill generation has 4 generation types (Off/Manual/Auto/Auto-Regen) and 5 schedule bases. Reconciliation has two flavors: penny-true-up inside allocation, and `rp_BillReconciliation` (Billed vs Calculated variance, source = interval or bill data). 14 reports depend on this subsystem.

Note: most allocation "methods" share one SQL engine — the UI precomputes the per-tenant percentages and writes them as method-1 fixed/percent rows. So to exercise each method we supply the right precomputed shares + method id.

Northlake today. The metering exists but nothing is configured — Cedar Row A and B are master-metered with an owned electric submeter and water submeter for each of the 124 apartments plus house submeters; there are no tenants and no allocation configs.

Gap. Entire subsystem absent.

What to add. A **mixed-use multi-tenant property** (proposed: Northlake Town Center — anchor retail + in-line shops + an office tower + a food court, and the already sub-metered Cedar Row apartments) structured to hit **every** method: fixed/%, fully & partially submetered, building-areas, coincident-demand, metered substations, and bill aggregation up to a cost center. Plus **WUI point families** (index + measured, with deliberate fail-reason cases), **generated tenant bills** under all 4 generation types, a **reconciliation** scenario with nonzero variance, and **tenant accruals**. This is also where the owner's "shopping mall / housing community" customers fit.

6. Accruals •

What EEMSuite supports. An accrual is a `Bill+BillItem` tagged with the special "Accruals" commodity (2300); its reversal carries "Reversals" (2301), paired in `Accrual_Bill/BillAccrualMap`. `rp_OpenAccruals` lists expected-but-unpaired accruals and can generate the negative reversal. "MPA" = **Massport** (a customer), an allocation/estimation generator — not "multi-point aggregate."

Northlake today. None.

What to add. Set the accrual/reversal commodity sys-constants; create an open accrual (a period missing its actual bill, with a 2300 line) and a trued-up accrual ↔ reversal pair; optionally an estimation generator run.

7. Bill validation ●

What EEMSuite supports. 47 QC rules (uaQC<n>_* procs) configured via `BillValidation` (per-account or global thresholds by measure type: % change, absolute floors/caps, period length). Modes: Bill Save (interactive) and Batch Test (job → report). A bill passes QC (status → 2) only with zero failures. ~30 of 47 rules early-return when there's no prior bill, so ≥2–3 chronological bills per account are required; two rules need a "Weather Normalized" scenario.

Northlake today. Deferred in Stage 1.

What to add. Validation thresholds per measure type; bills that deliberately pass and trip each rule family (cost/usage variance, period length, missing fields, inactive account); ≥1 open alert that blocks AP export; a "Weather Normalized" scenario for the weather-normalized rules.

8. GHG / emissions ●

What EEMSuite supports. `EmissionFactor(+History)` by emission type ($\text{CO}_2/\text{CH}_4/\text{N}_2\text{O}/\text{SF}_6/\text{HFCs}$... with GWP), `EmissionCategory` (scopes: indirect / stationary combustion / mobile / fugitive / process / de-minimis), `EmissionFactorSource(+Member)` (e.g. eGRID, EPA), mapped to meters (`EmissionMeterMap` / `EmissionFactorMeterMap`) and enabled per company (`EmissionCompanyConfig`). $\text{GHG} = \text{usage} \times \text{factor}$. Reports: `GHGHistory`, `GHGVariance`, `GHGAudit`.

Northlake today. Provider YAMLs mention eGRID/EPA factors, but nothing is configured in the EEMSuite emissions tables; meters aren't mapped to factors.

What to add. An emission factor source (eGRID region for Sacramento + EPA stationary combustion for gas), factor members with effective dates, meter → category mapping (electric = indirect/Scope 2; gas/steam = stationary combustion/Scope 1), and per-company scope config. Then GHG reports compute from the existing usage.

9. ENERGY STAR ratings ●

What EEMSuite supports. `ESFacility` (PrimaryFunction, GFA, occupancy, PMPropertyID) + rating **requests** (`ESRequest`) and **responses/scores** (`ESResponseFacility`). Reports: `RatingComparison`, `RatingHistory`, `RequestLog`. Scores normally come from the Portfolio Manager API.

Northlake today. Excellent property mapping (13 properties, exact PrimaryFunction enums, use details) — but **no rating requests or responses**, so the rating reports are empty.

What to add. Seed `ESResponseFacility` scores (and request/log rows) directly for the mapped properties across several periods, so `RatingHistory/Comparison` populate without a live PM call.

10. Weather regression / normalization •

What EEMSuite supports. Regression of use vs HDD/CDD (+ optional 3rd variable) stored in FF_wnRegressionResults / FF_wnRegressionBestResults (intercept, HDD/CDD factors, R^2 , CV-RMSE, t-stats), produced by a regression **task** — not by raw data load. Drives WeatherRegression, RegressionSummary, and the weather-normalized variants of ScatterPlot/UsageVariance.

Northlake today. HDD/CDD related points + 3 baseline shells exist; **no stored regression results.**

What to add. Run/seed regression results for the metered buildings (using the existing interval + weather data) so the regression and weather-normalized reports return models with realistic fit stats.

11. Benchmarks / EUI •

What EEMSuite supports. wh_BenchMark + EUI normalization (use ÷ floor area), peer sets, performance snapshot; relates to ENERGY STAR score. Reports: BenchmarkRanking, MonthlyEUI, PerformanceSnapshot.

Northlake today. Floor areas exist (so EUI is computable); no benchmark definition or peer grouping.

What to add. Benchmark definitions (EUI, cost intensity), and enough comparable buildings/periods that ranking is meaningful (the campus buildings + town-center + Cedar Row give a peer set).

12. Project tracking •

What EEMSuite supports. Project (type/category, capital/operating cost, incentives) → Activity (design/install/M&V) → Event (milestones), tied to hierarchy; M&V activities relate baselines to post-project use. Reports: ProjectDetail, ProjectSummary, ProjectFinance.

Northlake today. None.

What to add. A small project portfolio (e.g. LED retrofit, chiller upgrade, solar expansion) with activities, financial events, and M&V linkage to the baselines — realistic for a campus sustainability program.

13. Alarms •

What EEMSuite supports. AlarmData_TBL raised on points, with PriorityLevels (informational → emergency) and alarm types. Report: AlarmDataMap.

Northlake today. None.

What to add. Alarm-enabled points (e.g. demand threshold, chiller fault, data-gap) with raised + acknowledged + resolved alarm rows across the data span.

14. Audit / activity logs ●

What EEMSuite supports. AuditLog/AuditConfig (data-change audit, must be enabled) and user-activity tracking. Reports: Audit, ActivityTracker.

Northlake today. None.

What to add. Enable audit; generate a realistic trail of user actions and data edits across the 18+ users (and the disabled-user negative case).

15. Tasks / job scheduler / workflows ●

What EEMSuite supports. A task engine with **6 task types** (System, Standard Alerts, EEM Gateways, Report Tasks, MFR Reports, Bill Importer) plus a component-based **workflow** engine (gateways, REST, SQL, file transfer, email, run-process, control-flow nodes), scheduled by interval/cron. TaskLog records runs. Reports: TaskLog, GatewayStatus, job_scheduler dashboard.

Northlake today. 15 gateway **profiles** are designed, but there is no task run history, no schedules, no workflows.

What to add. Task schedules for the gateways + bill import + report delivery; a couple of multi-step workflows (e.g. download → bill-import → validate → notify; gateway → interval publish → regression → alert); and TaskLog history so TaskLog/GatewayStatus report real runs.

16. MFR saved reports & delivery (100+ users) ●

What EEMSuite supports. Users save parameterized reports (GUIMFRSavedReports_TBL + parameters), build dashboards (GUIMFRDashboardReports), and schedule recurring **delivery** (MFRDelivery_TBL → MFRDeliveryReport_TBL + MFRDeliveryRecipient_TBL) to individuals, **groups**, or external emails, rendered as PDF/Excel by a delivery task. **92 of 115 reports are MFR-deliverable.**

Northlake today. None.

What to add. A library of saved reports + dashboards; delivery schedules whose recipient lists resolve to **100+ users** (the owner's explicit target) — most efficiently via **group** recipients in a large portfolio company, plus a few individual and external recipients; the delivery task to execute them. This needs the **scale** decision (how many users/companies).

17. Importers / file intake ●

What EEMSuite supports. Bill Importer (BIF + CSV/fixed-width/Excel, with source/utility/account/commodity mapping + validation), GreenButton/ESPI XML interval import, MyImportExport templates, and MyUpload (JSON/CSV) handlers. Reports: bill_importer, my_imports, my_exports.

Northlake today. Handheld (Neptune/MVRS) and gateway files are designed; **no bill-importer source mappings, no GreenButton XML, no MyUpload/MyImport configs.**

What to add. A bill-importer source + mapping + sample BIF/CSV/Excel files (incl. the shipped bad-data validation sample), a GreenButton ESPI XML feed for one meter, and a MyUpload/MyImport template — so each intake path is exercised alongside the gateways.

18. HMR (handheld meter reading) ●

What EEMSuite supports. HMR_Meter (route/cycle/sequence), HMR_ReadType (dials/decimals/direction/demand), HMR_Reading. Reports: HMRConfiguration, hmr_export.

Northlake today. Neptune (water) + MVRS (gas) upload files are designed.

What to add. Explicit HMR meter/route/read-type config so HMRConfiguration renders and route uploads tie to HMR readings.

19. EnergyAI ●

What EEMSuite supports. Enroll a load point + baseline period; push consumption + weather to the EnergyAI service; store returned insights in EnergyAI_Insights. Report: EnergyAIInsights.

Northlake today. None.

What to add. Pre-generated insight rows for a couple of well-instrumented load points (anomalies, forecasts) so the report renders without a live service.

20. Commodities & unique units ● (extensible)

What EEMSuite supports. 37 measure types (electricity, gas, steam, chilled water, potable/sewer water, fuel oils, propane, diesel, kerosene, LPG, hot water, coal, gasoline, recycling, solid waste, NGL, ethane, oil, produced water, plus money/weather/activity/digital/sensor/emission). Rich unit model with conversions, compound/rate units, dynamic (time-based) conversions, multi-currency, and unit profiles per user.

Northlake today. Uses ~9 commodities with appropriate units including the unusual ones (ton-hours, klb steam, ERU). Good.

What to add (optional breadth). A fuel-oil or propane backup-boiler meter, a diesel generator, and a multi-currency rate would extend unit coverage; a custom user-defined measure type proves the "unique units" path.